CS2033: Data Communication and Networking

Data Transmission

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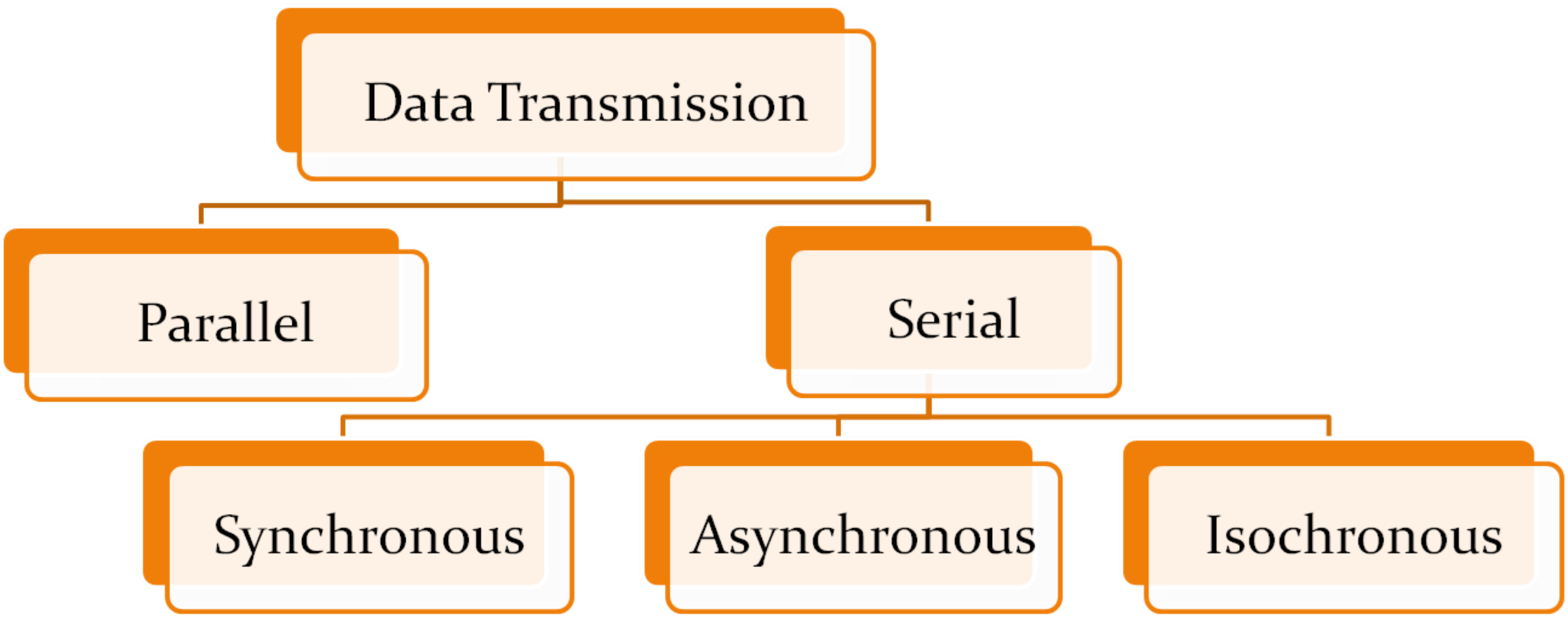
Outline

- Transmission modes
  - Parallel
  - Serial
- Multiplexing
  - Frequency Division Multiplexing
  - Synchronous Time Division Multiplexing
  - Statistical Time Division Multiplexing
- Implementation example: ADSL

Transmission Modes

- The way in which a bit group goes from one device to another



Parallel Transmission

- Parallel: group of bits sent simultaneously by using a separate line for each bit
- Pros – Fast
- Cons - Over longer distances;
  - costly to use many wires
  - need thicker wires to reduce signal degradation and bundling into a single cable is not practical
  - bits may not be received simultaneously

Serial Transmission

- Serial: bits sent over a single line one bit after another

- Pros:
  - Cheap
  - Reliable
- Cons:
  - Complexity because, sender and receiver must determine the order of bits

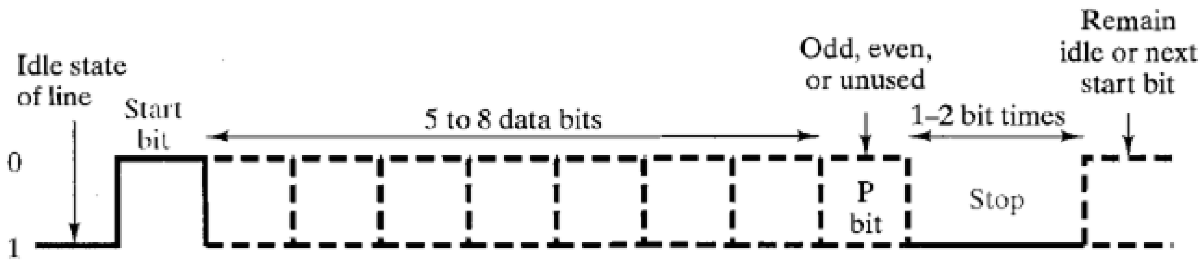
Asynchronous Transmission

- Bits are divided into small groups (e.g. byte) and sent independently
  - E.g. keyboard inputs
- Properties of the communication
  - receiver does not know when a group of bits will arrive
  - unpredictable time intervals between transmissions
  - timing maintained within each group

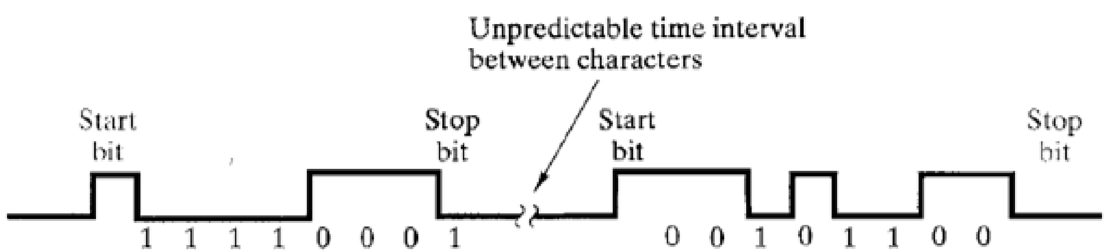
Asynchronous Transmission

- Must signal that sender is going to send a group of bits before sending actual bits
  - Start Bit – changes the line from idle state (normally considered as binary 1) to 0
- Must signal that sender has finished sending a group of bits
  - Stop Bits – changing line state back to logic 1 (idle)
- Cons – Too much of overhead bits

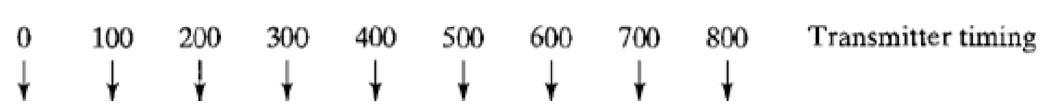
Example for Asynchronous Tx



(a) Character format



(b) 8-bit asynchronous character stream



Synchronous Transmission

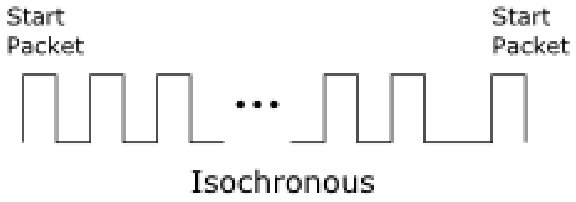
- Larger bit groups are sent without start/stop bits for each character
- Such a bit group is called a Frame
- Frame structure varies from protocol to protocol
- Organization of a generic frame

Syn	Ctrl	Data	Error	End

Syn – synchronization bits
Ctrl – control bits
Data – Data bits
Error – Error checking bits
End – End-of-frame bits

Isochronous Transmission

- Combines features of both synchronous and asynchronous
- An isochronous data transfer system sends blocks of data asynchronously, in other words the data stream can be transferred at random intervals.
- Each transmission begins with a start packet. Once the start packet is transmitted, the data must be delivered with a guaranteed bandwidth. Isochronous data transfer is commonly used for where data must be delivered within certain time constraints, like streaming video.



Source: http://bucarotechelp.com/networking/basics/88092302.asp

Simplex, half duplex and duplex

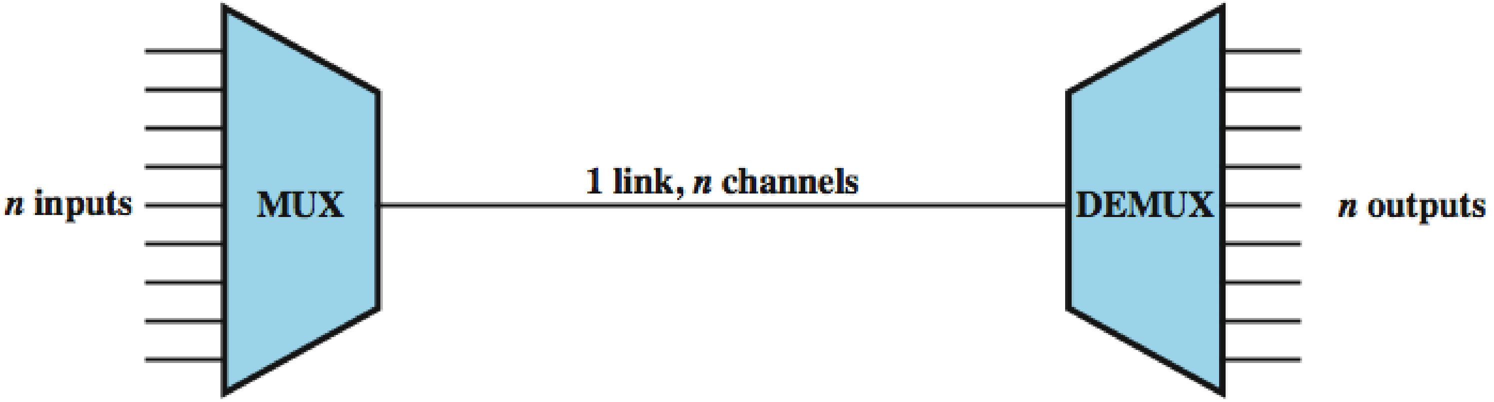
- simplex
  - one direction
    - eg. television
- half duplex
  - either direction, but only one way at a time
    - eg. police radio
- full duplex
  - both directions at the same time
    - eg. telephone

Multiplexing

- Many to one!
- Multiplexer (Mux): device that routes transmission from multiple sources to a single destination
- Frequency Division Multiplexing – e.g. Television
- Time Division Multiplexing – e.g. digitized data transmission
  - Synchronous TDM
  - Statistical TDM

Multiplexing

- multiple links on 1 physical line
- common on long-haul, high capacity, links
- have FDM, TDM, STDM alternatives



Why multiplexing is popular?

- The higher the data rate, the more cost-effective the transmission facility.

- For a given application and over a given distance, the cost per kbps declines with an increase in the data rate. Similarly, the cost of transmission and receiving equipment, per kbps, declines with increasing data rate.

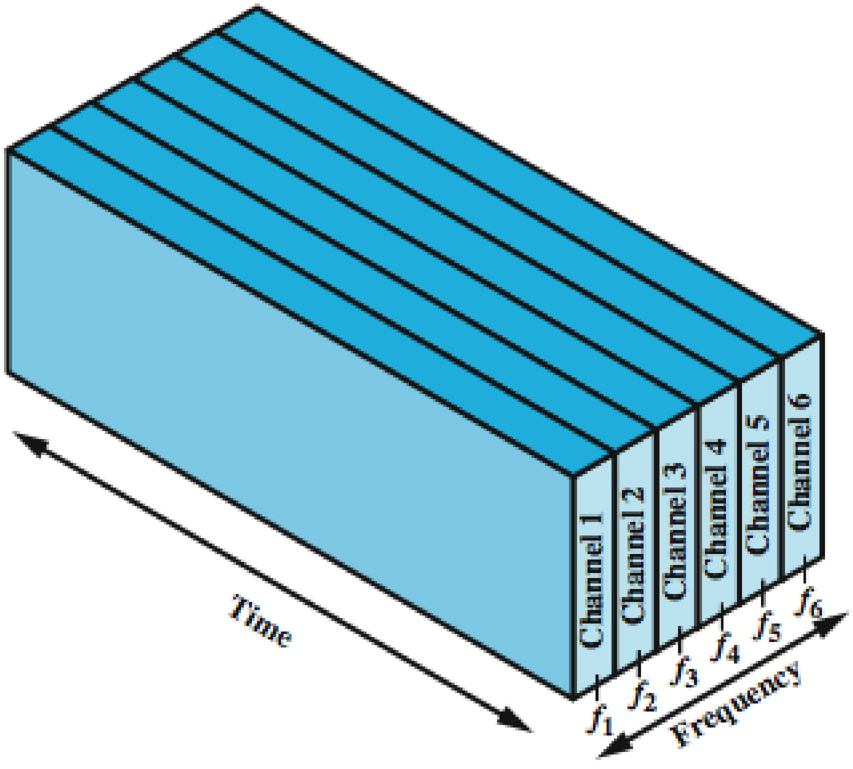
- Most individual data communicating devices require relatively modest data rate support.

Frequency Division Multiplexing (FDM)

- FDM is possible when the useful bandwidth of the transmission medium exceeds the required bandwidth of signals to be transmitted.

- Multiple signals can be carried simultaneously if each signal is modulated onto a different carrier frequency and the carrier frequencies are sufficiently separated that the bandwidths of the signals do not significantly overlap.

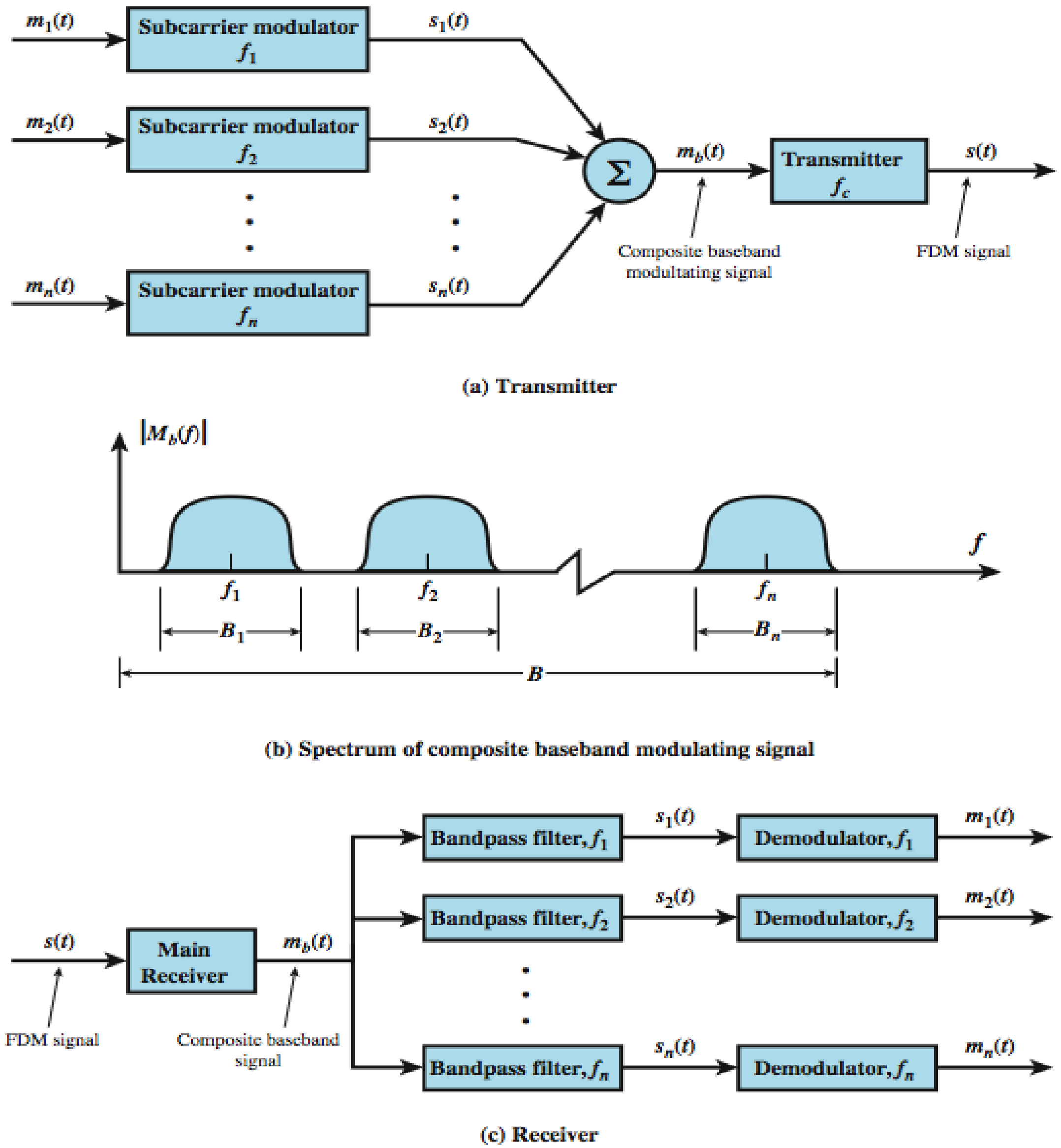
Frequency Division Multiplexing (FDM)



(a) Frequency division multiplexing

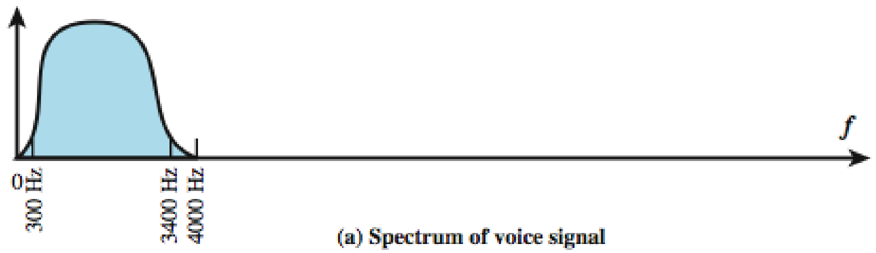
FDM Process

1. A number of analog or digital signals are to be multiplexed onto the same transmission medium.
2. Each signal is modulated onto a carrier – referred to as a subcarrier. Any type of modulation may be used.
3. Resulting analog, modulated signals are then summed to produce a composite baseband signal

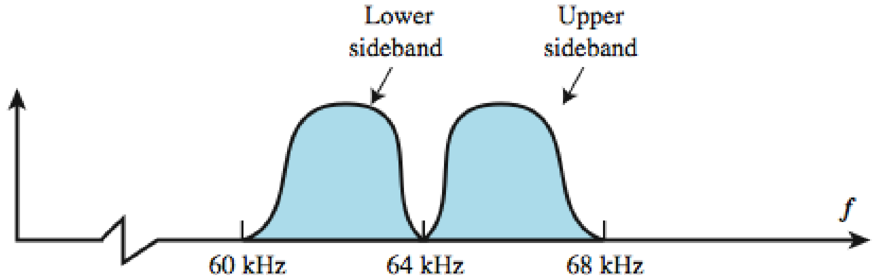


FDM Example

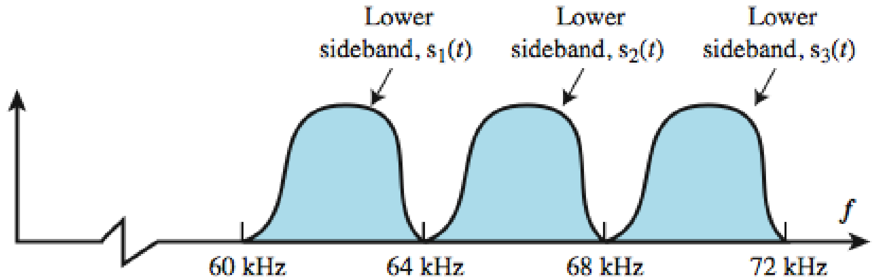
- Three voice signals need to be sent simultaneously
- Voice signal bandwidth is usually considered as 4kHz. (effective spectrum from 300Hz to 3400Hz)



(a) Spectrum of voice signal



(b) Spectrum of voice signal modulated on 64 kHz frequency

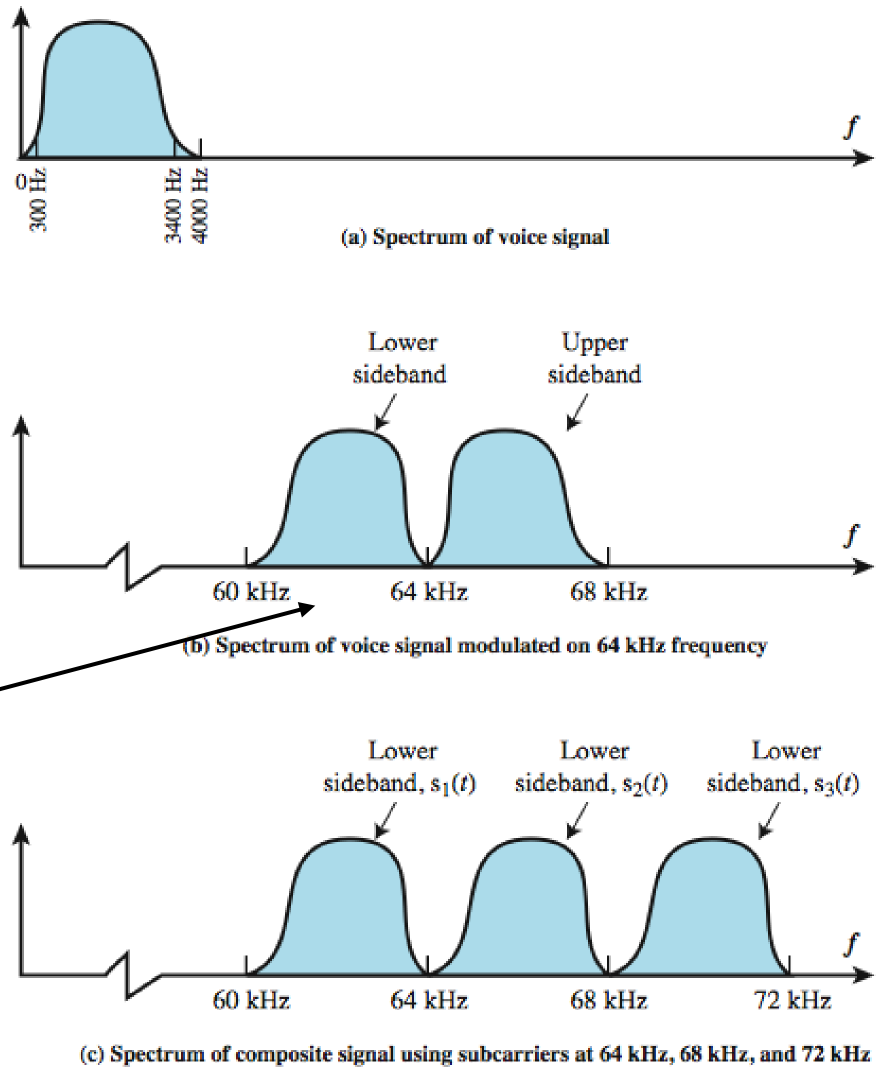


(c) Spectrum of composite signal using subcarriers at 64 kHz, 68 kHz, and 72 kHz

FDM Example

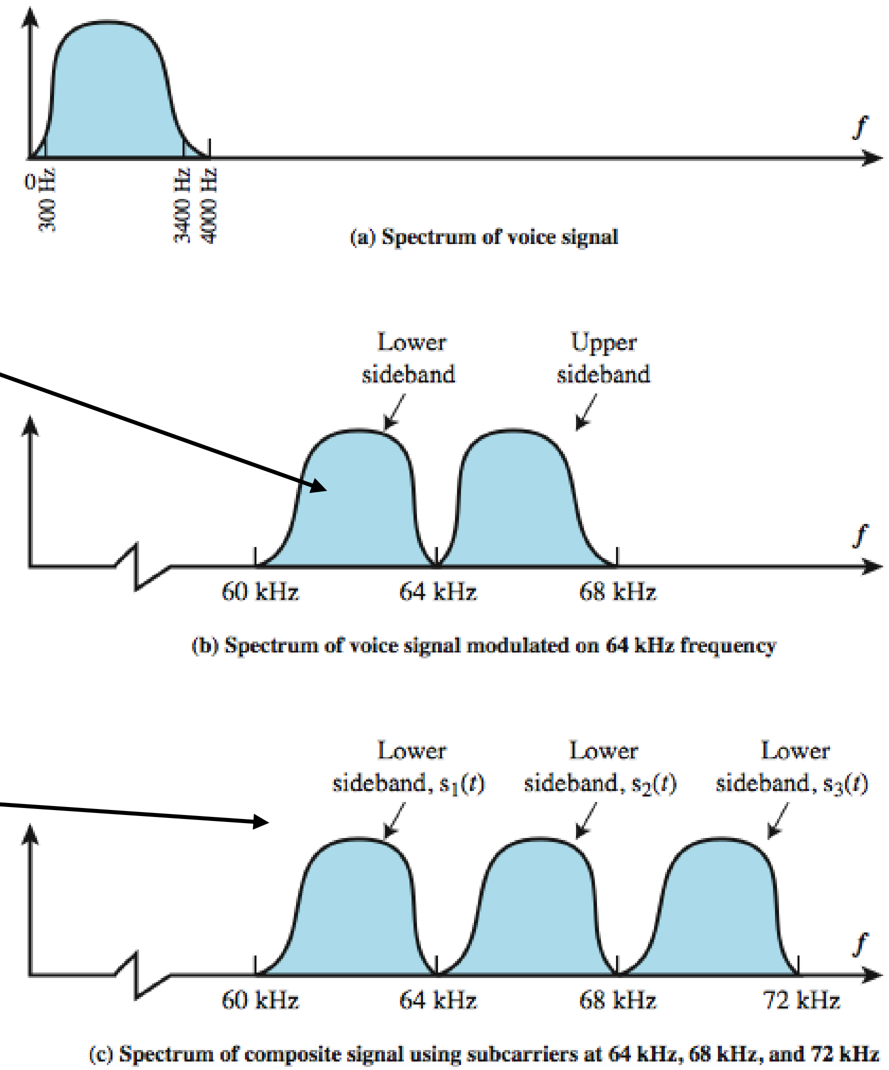
- Suppose voice signal is amplitude modulated with a 64kHz carrier.

- Modulated signal has a bandwidth of 8kHz (with two sidebands)



FDM Example

- To make efficient use of bandwidth we can transmit only one sideband. (let's say lower one is picked)
- Can send lower sideband of 3 voice signals modulated carriers 64, 68 and 72kHz simultaneously.



FDM Issues

- Crosstalk: May occur if the spectra of adjacent component signals overlap significantly

- Intermodulation noise: On a long link, the nonlinear effects of amplifiers on a signal in one channel could produce frequency components in other channels.

Can we use FDM for Fiber Optics?

- True potential of optical fiber is fully exploited when multiple beams of light at different frequencies are transmitted on the same fiber.

- Form of frequency division multiplexing (FDM) but is commonly called **wavelength division multiplexing (WDM).**

- With WDM, the light streaming through the fiber consists of many colors, or wavelengths, each carrying a separate channel of data.

Wavelength Division Multiplexing

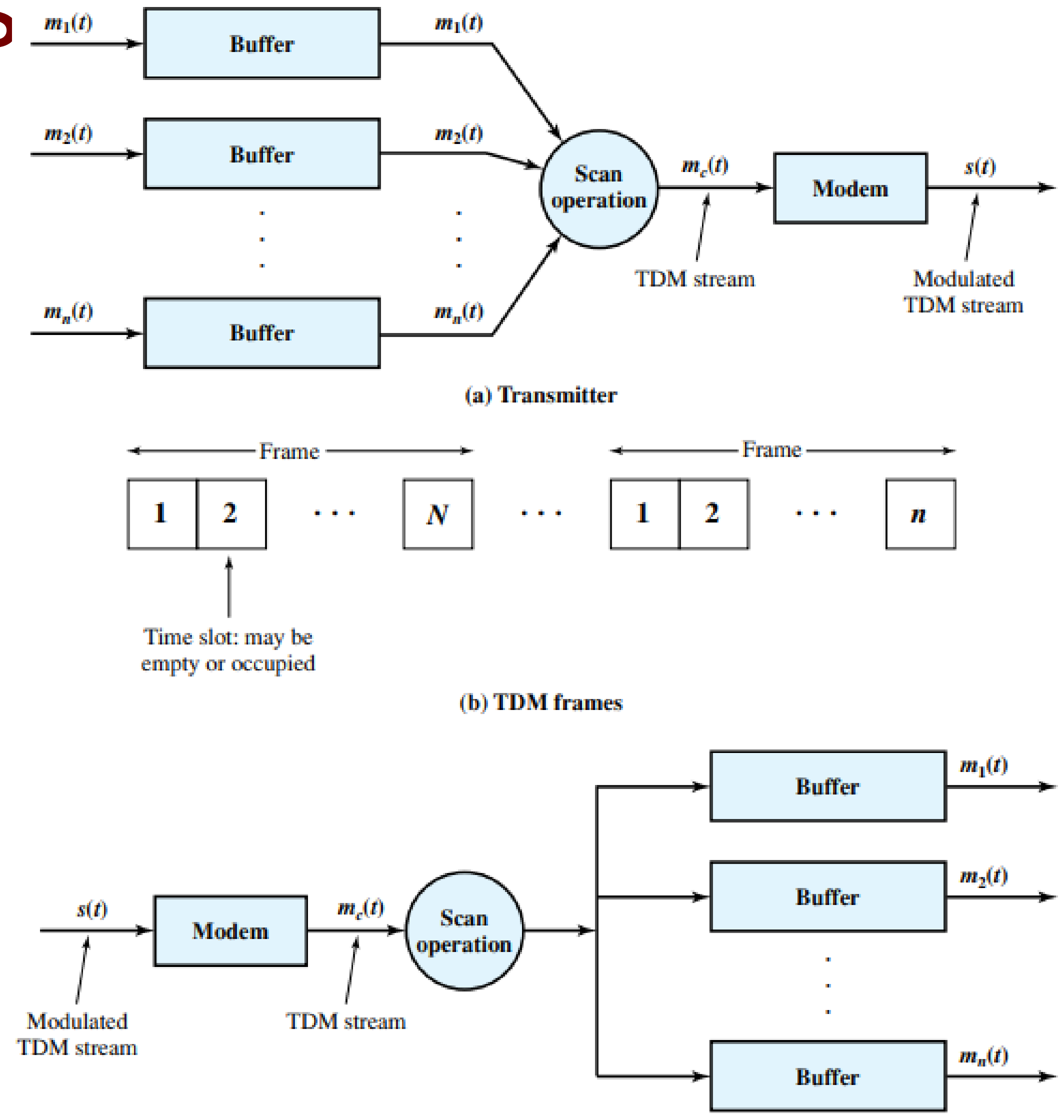
- Several sources generate a laser beam at different wavelengths.
- These are sent to a multiplexer, which consolidates the sources for transmission over a single fiber line.
- Optical amplifiers, typically spaced tens of kilometers apart, amplify all of the wavelengths simultaneously.
- Finally, the composite signal arrives at a demultiplexer, where the component channels are separated and sent to receivers at the destination point.

Synchronous Time Division Multiplexing (TDM)

- Synchronous time division multiplexing is possible when the achievable data rate of the medium exceeds the data rate of digital signals to be transmitted.
- Multiple digital signals (or analog signals carrying digital data) can be carried on a single transmission path by interleaving portions of each signal in time.
- The interleaving can be at the bit level or in blocks of bytes or larger quantities.

TDM Process

1. A number of signals are to be multiplexed onto the same transmission medium. The signals carry digital data and are generally digital signals.
2. The incoming data from each source are briefly buffered. Each buffer is typically one bit or one character in length. The buffers are scanned sequentially to form a composite digital data stream

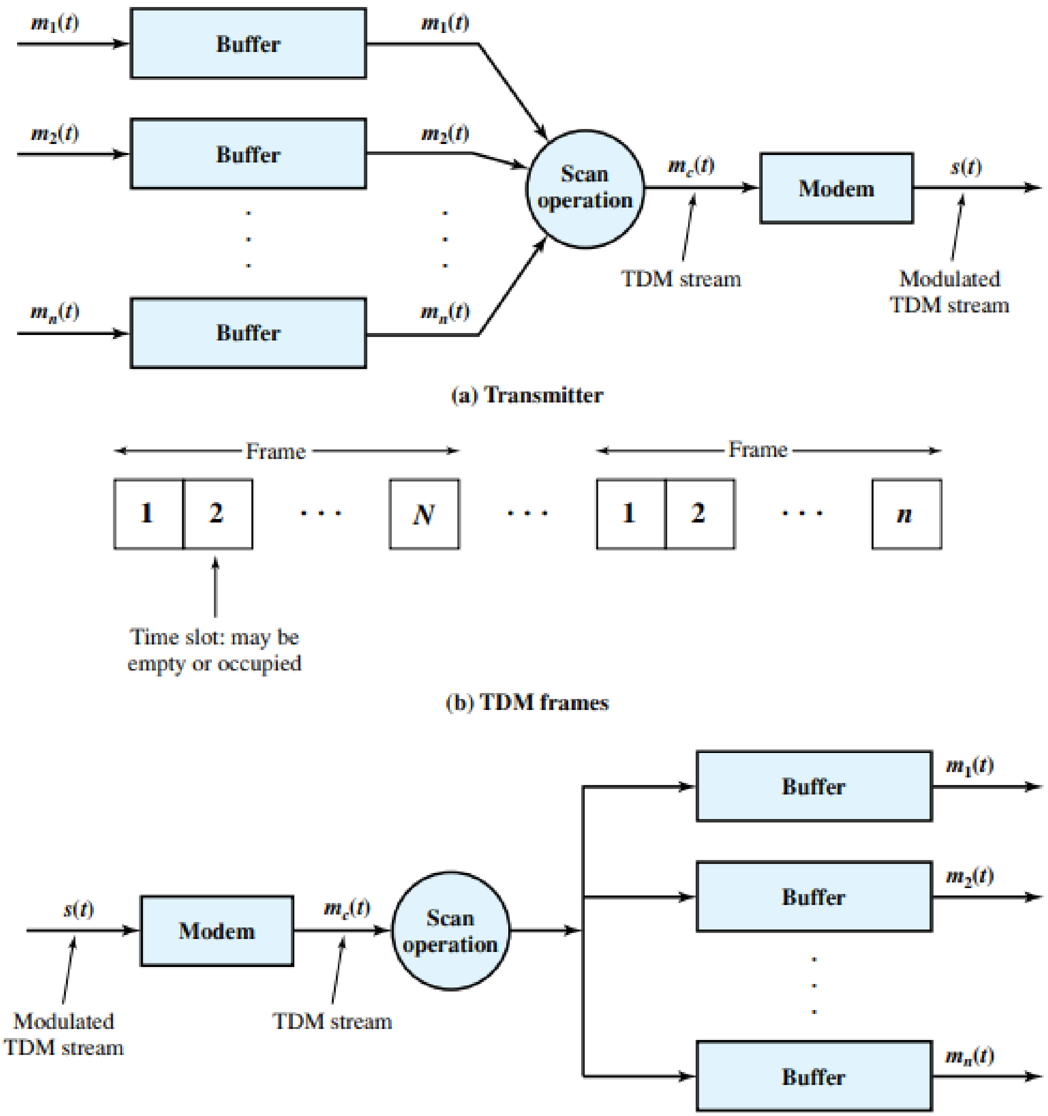


TDM Process

3. The data are organized into frames. Each frame contains a cycle of time slots.

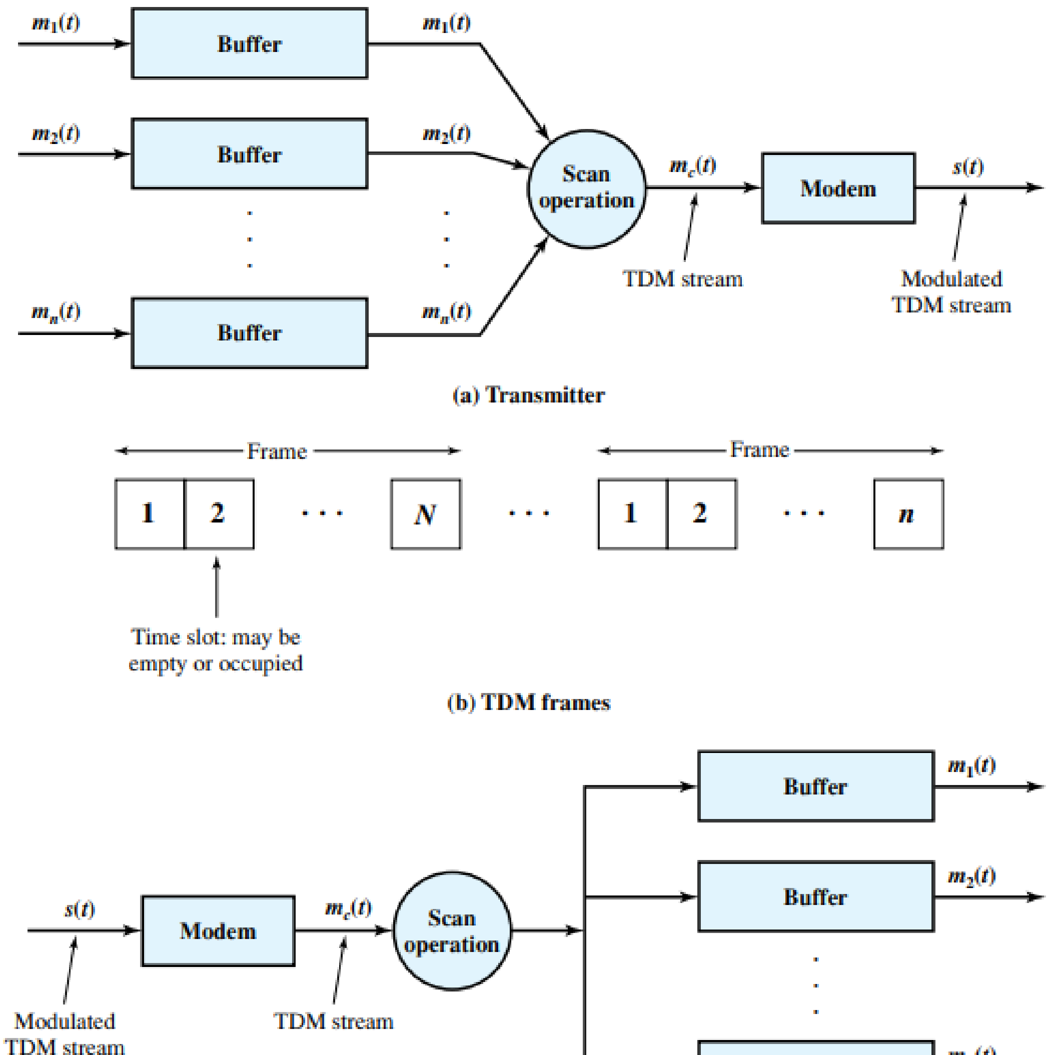
   In each frame, one or more slots are dedicated to each data source. The sequence of slots dedicated to one source, from frame to frame, is called a channel.

   The slot length equals the transmitter buffer length, typically a bit or a byte (character).



TDM Process

4. At the receiver, the interleaved data are demultiplexed and routed to the appropriate destination buffer.



Synchronous TDM

- Synchronous TDM is called synchronous not because synchronous transmission is used, but because the time slots are preassigned to sources and fixed. The time slots for each source are transmitted whether or not the source has data to send.
- This is the case with FDM as well. Capacity is wasted to achieve simplicity of implementation.
- Even when fixed assignment is used, however, it is possible for a synchronous TDM device to handle sources of different data rates. For example, the slowest input device could be assigned one slot per cycle, while faster devices are assigned multiple slots per cycle.

TDM Link Control

- Flow control mechanisms are not needed as far as TDM mux and demux are concerned. The data rate on the multiplexed line is fixed, and the multiplexer and demultiplexer are designed to operate at that rate.

- But what if one of the devices attached to an output line is unable to accept data? Should TDM transmission stop?

  No! Affected channel will carry empty slots and others, data as usual.

TDM Link Control

- For error control also, it would not request retransmission of an entire TDM frame because an error occurs on one channel. Error control will be handled on a per-channel basis.

- For flow and error control a data link control protocol such as HDLC (High-Level Data Link Control) can be used.

TDM Link Control with HDLC

(a) Configuration

$\text{Input}_1 \cdots\cdots\cdots\; F_1\ f_1\ f_1\ d_1\ d_1\ d_1\ C_1\ A_1\ F_1\ f_1\ f_1\ d_1\ d_1\ d_1\ C_1\ A_1\ F_1$

$\text{Input}_2 \cdots\; F_2\ f_2\ f_2\ d_2\ d_2\ d_2\ d_2\ C_2\ A_2\ F_2\ f_2\ f_2\ d_2\ d_2\ d_2\ d_2\ C_2\ A_2\ F_2$

(b) Input data streams

$\cdots f_2\ F_1\ d_2\ f_1\ d_2\ f_1\ d_2\ d_1\ d_2\ d_1\ C_2\ d_1\ A_2\ C_1\ F_2\ A_1\ f_2\ F_1\ f_2\ f_1\ d_2\ f_1\ d_2\ d_1\ d_2\ d_1\ d_2\ d_1\ C_2\ C_1\ A_2\ A_1\ F_2\ F_1$

(c) Multiplexed data stream

Legend: F = flag field d = one octet of data field
A = address field f = one octet of FCS field
C = control field

TDM Framing

- Because we are not providing flag or SYNC characters to bracket TDM frames, some means is needed to assure frame synchronization.

- An identifiable pattern of bits, from frame to frame, can be used as a “control channel”.

- Example: alternating bit pattern, 101010. This is a pattern unlikely to be sustained on a data channel.

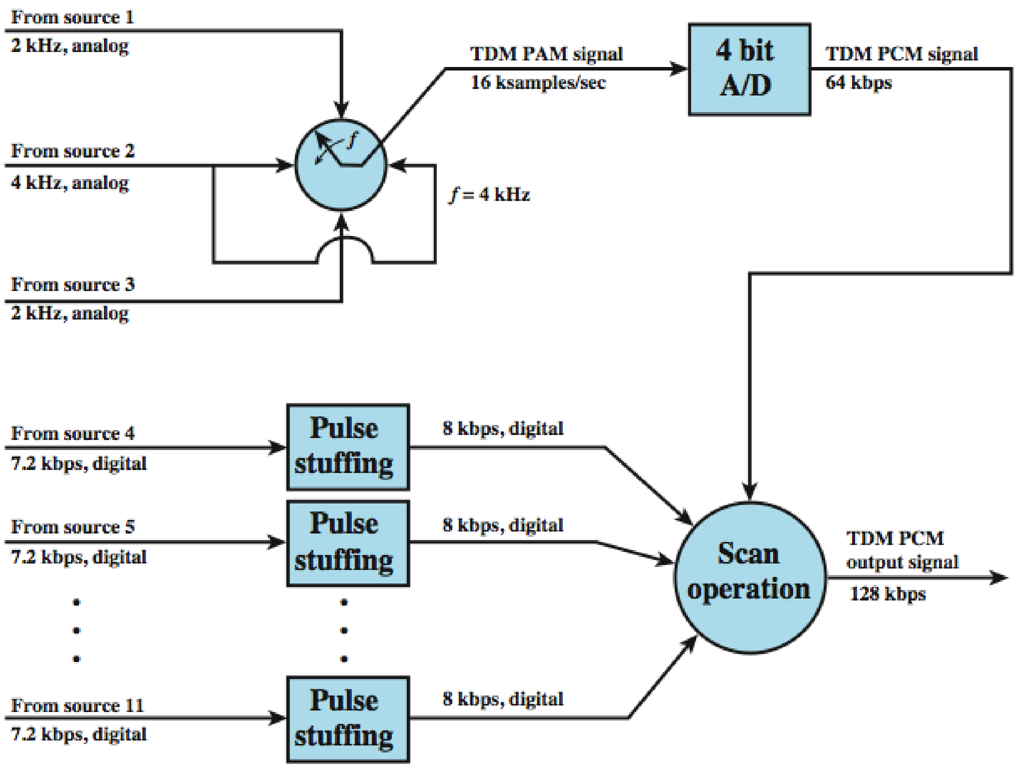
Pulse Stuffing

- Need of synchronizing data sources due clocks in different sources drifting
- Data rates from different sources not equal or having a simple relation
- Pulse Stuffing a common solution
  - have outgoing data rate (excluding framing bits) higher than sum of incoming rates
  - stuff extra dummy bits or pulses into each incoming signal until it matches local clock
  - stuffed pulses inserted at fixed locations in frame and removed at demultiplexer

TDM Example

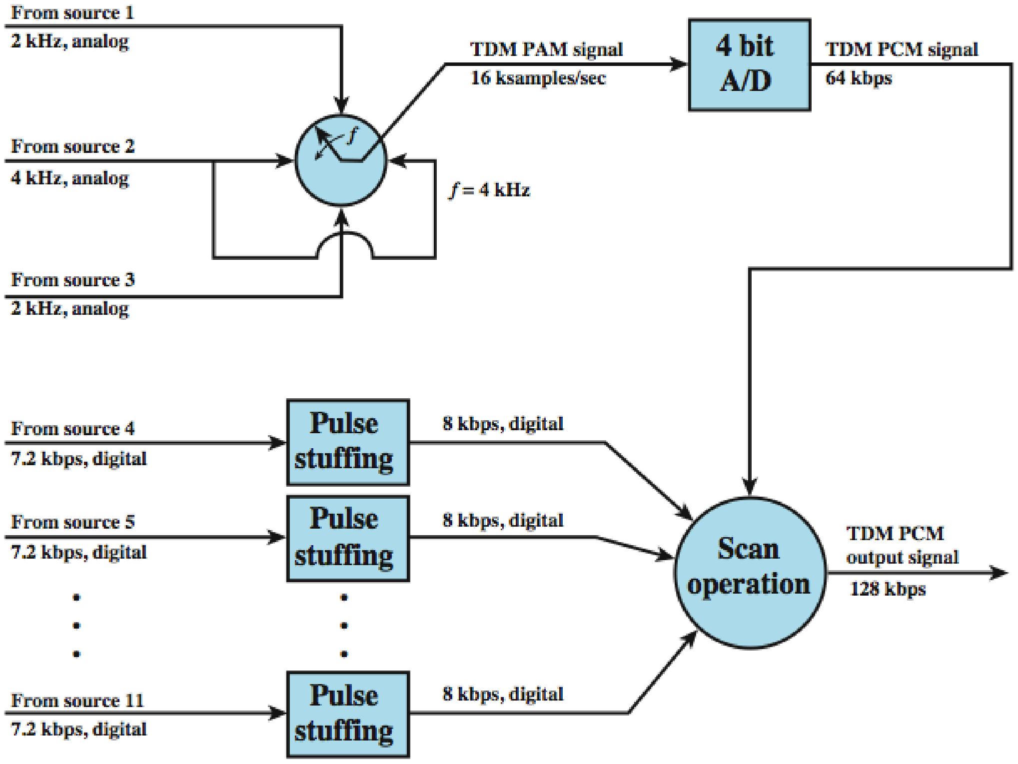
- Consider that there are 11 sources to be multiplexed on a single link.

- The analog sources are converted to digital using PCM.



TDM Example

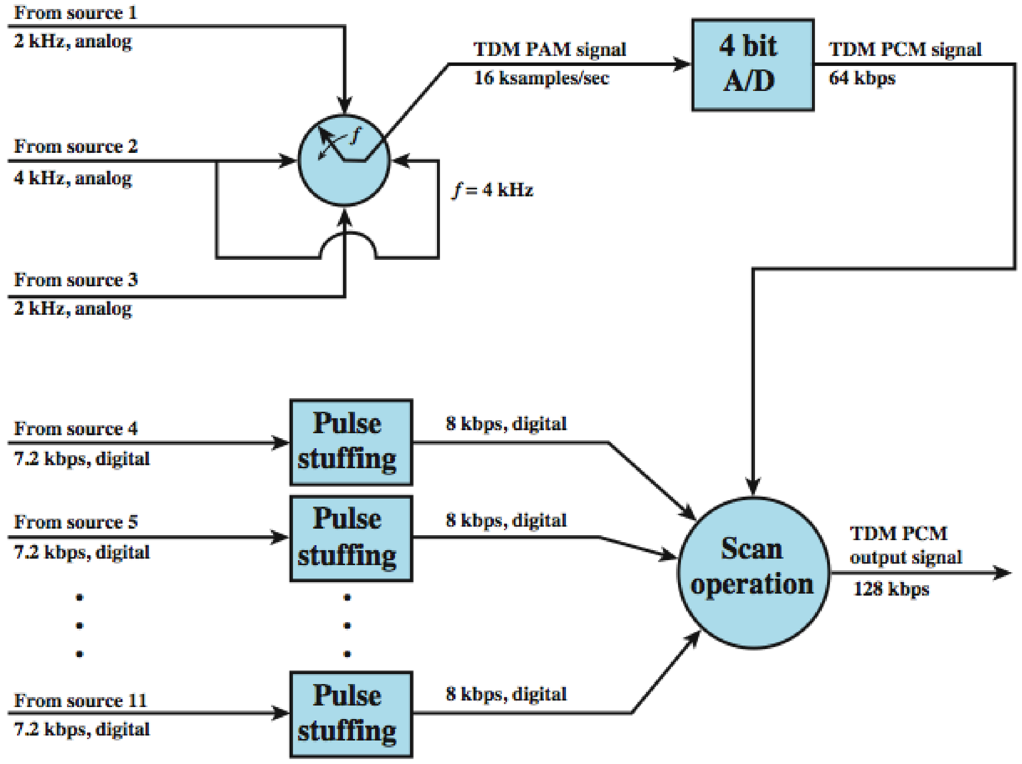
- PCM is based on the sampling theorem, which dictates that a signal be sampled at a rate equal to twice its bandwidth.
- Thus, the required sampling rate is 4000 samples per second for sources 1 and 3, and 8000 samples per second for source 2.



TDM Example

- These samples, which are analog, must then be quantized or digitized. Let us assume that 4 bits are used for each analog sample.

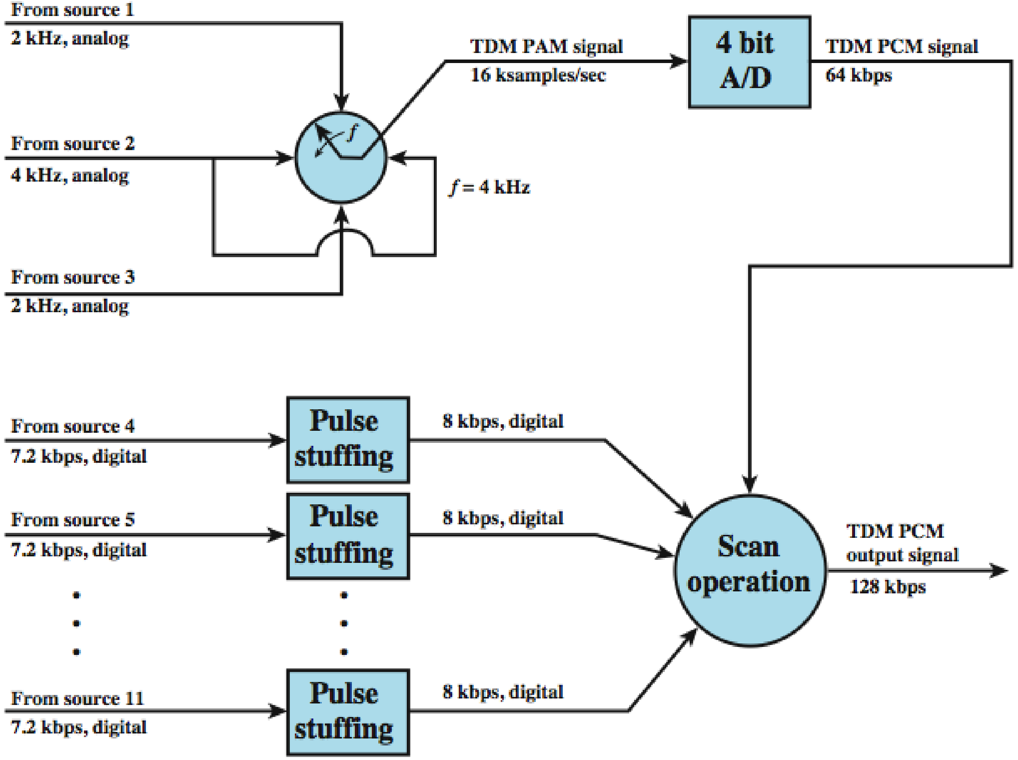
- For convenience, these three sources will be multiplexed first, as a unit.



TDM Example

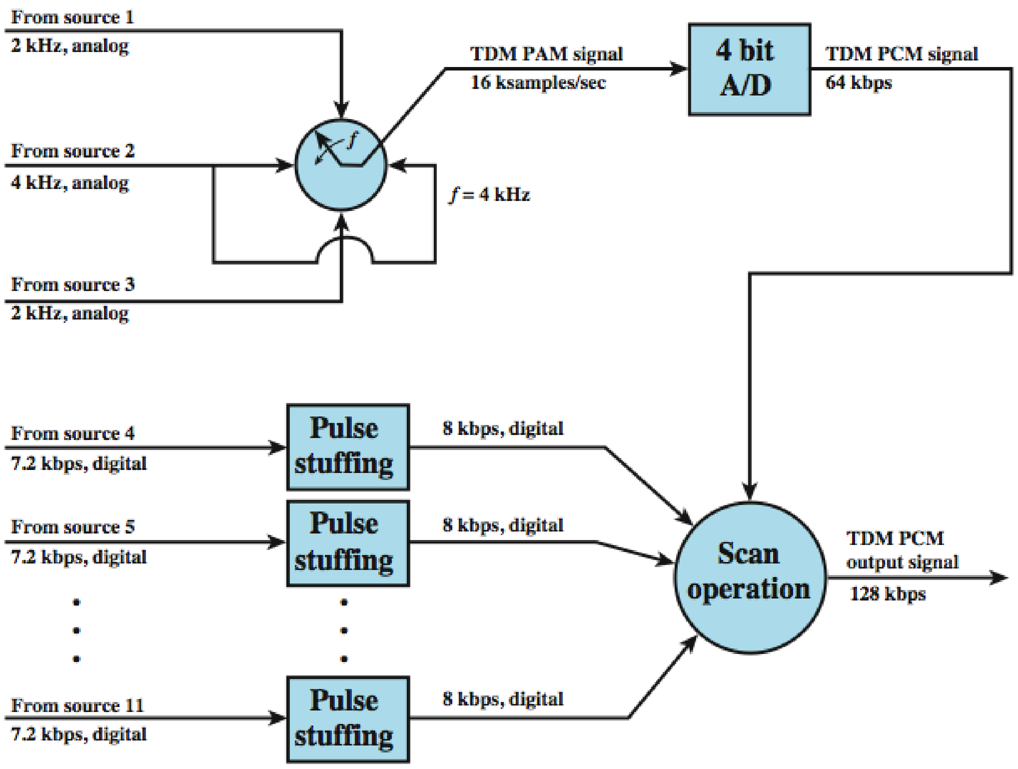
- At a scan rate of 4 kHz, one sample each is taken from sources 1 and 3, and two samples are taken from source 2 per scan.

- These four samples are interleaved and converted to 4-bit PCM samples. Thus, a total of 16 bits is generated at a rate of 4000 times per second, for a composite bit rate of 64 kbps.



TDM Example

- For the digital sources, pulse stuffing is used to raise each source to a rate of 8 kbps, for an aggregate data rate of 64 kbps.



Statistical TDM

- In a synchronous time division multiplexer, it is often the case that many of the time slots in a frame are wasted. A typical application of a synchronous TDM involves linking a number of terminals to a shared computer port. Even if all terminals are actively in use, most of the time there is no data transfer at any particular terminal.

- An alternative to synchronous TDM is statistical TDM

Statistical TDM

- The statistical multiplexer exploits this common property of data transmission by dynamically allocating time slots on demand.

- In the case of the statistical multiplexer, there are n I/O lines, but only k, where k<n, time slots available on the TDM frame.

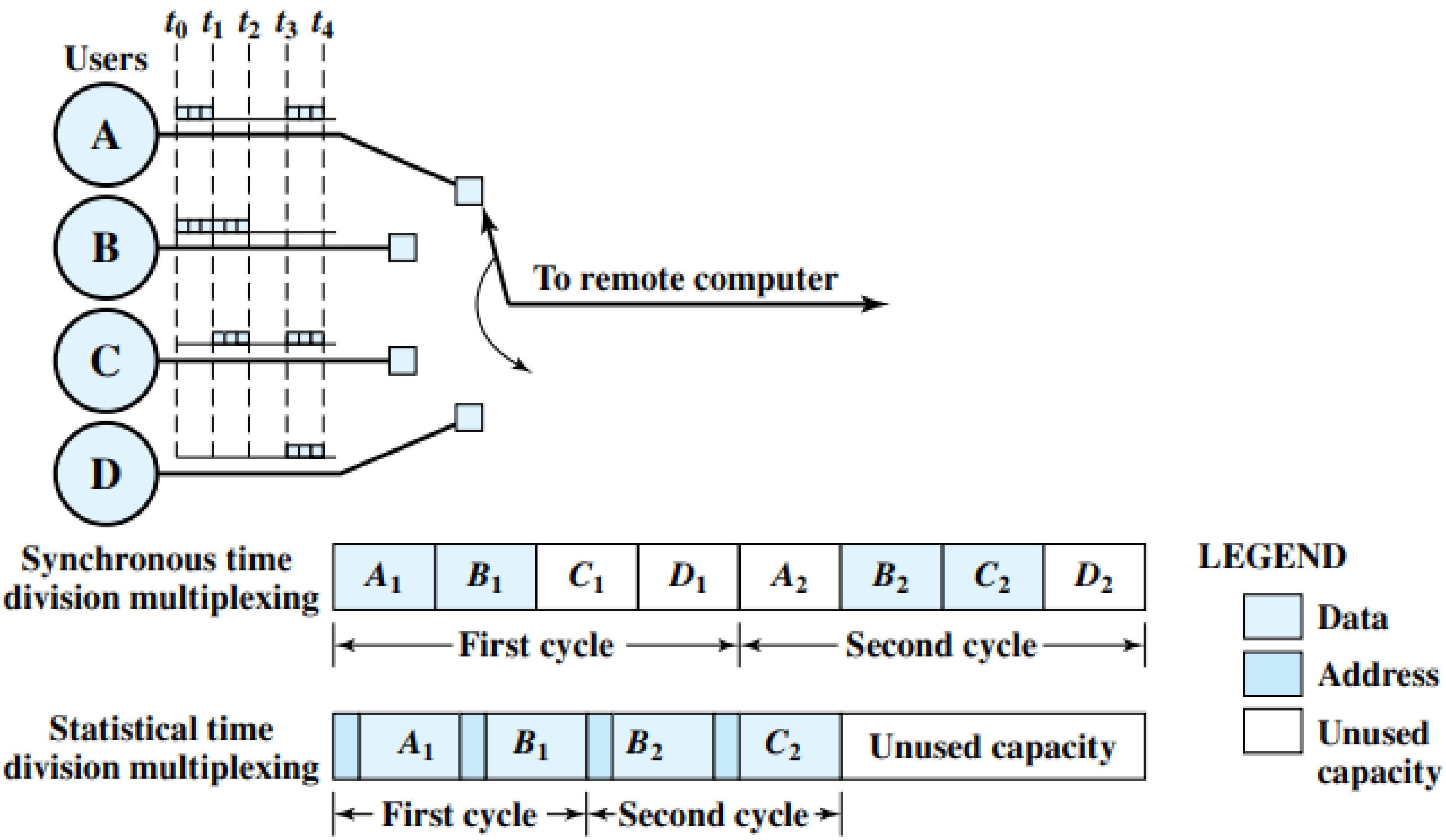
- Multiplexer: scan the input buffers, collecting data until a frame is filled, and then send the frame.
- Demultiplexer: receives a frame and distributes the slots of data to the appropriate output buffers.

Statistical TDM

- A statistical multiplexer can use a lower data rate to support as many devices as a synchronous multiplexer.

- Alternatively, if a statistical multiplexer and a synchronous multiplexer both use a link of the same data rate, the statistical multiplexer can support more devices.

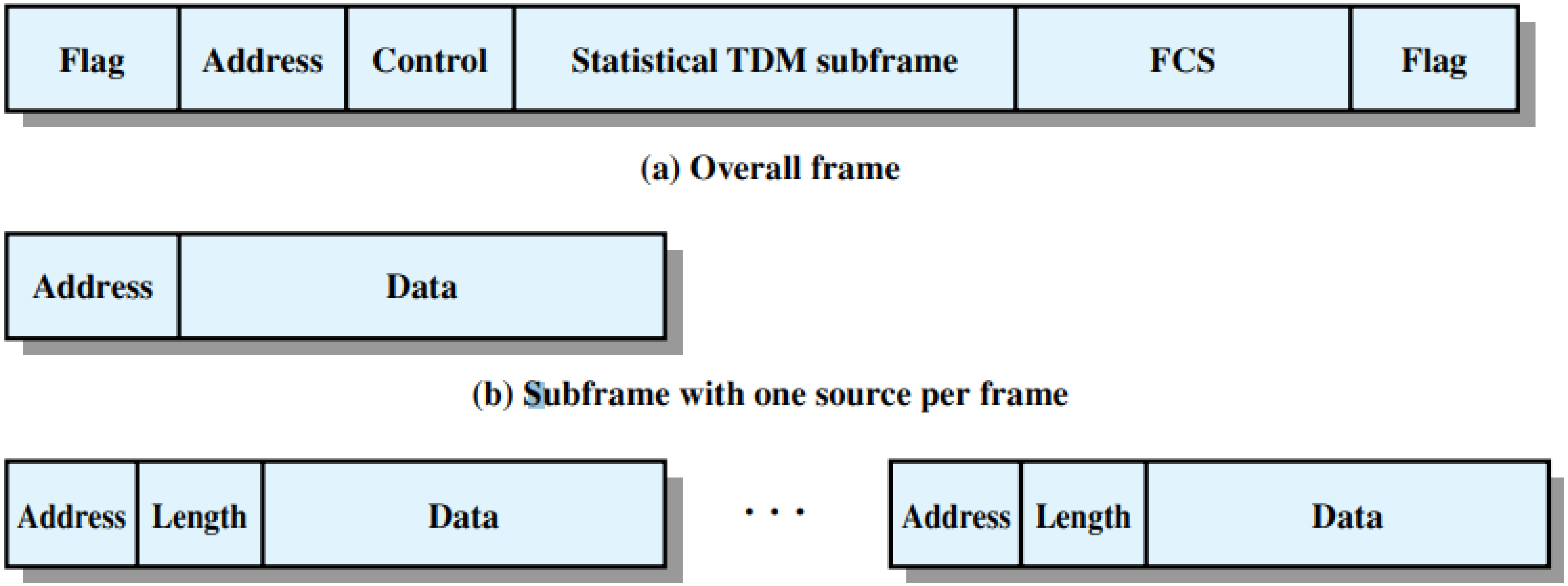
Synchronous vs Statistical TDM



Frame Structure

- Typically, a statistical TDM system will use a synchronous protocol such as HDLC.
- Within the HDLC frame, the data frame must contain control bits for the multiplexing.
- The frame structure can be optimized to improve performance.

Frame Structure



(a) Overall frame

(b) Subframe with one source per frame

(c) Subframe with multiple sources per frame

Asymmetric Digital Subscriber Line (ADSL)

- Makes use of existing voice telephone network to create a digital subscriber line link between the network and subscriber.

- Voice lines were installed to carry 0-4kHz, but are capable of handling more than 1Mhz spectrum.

- ADSL makes use of these voice lines for creating a digital subscriber link.

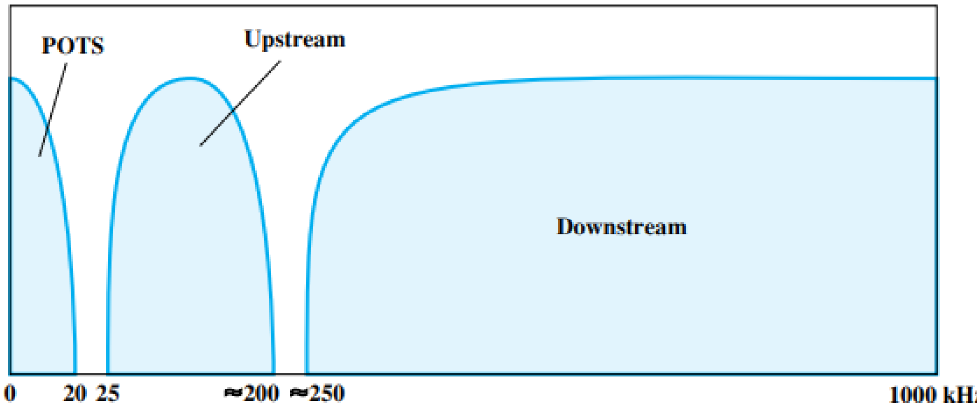
Asymmetric Digital Subscriber Line (ADSL)

- The term asymmetric refers to the fact that ADSL provides more capacity downstream than upstream.

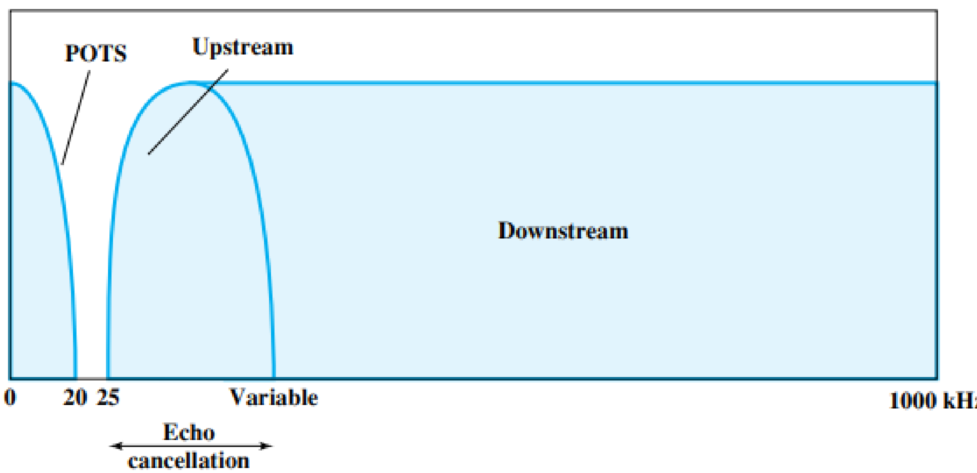
- ADSL uses frequency division multiplexing (FDM) in a novel way to exploit the 1-MHz capacity of twisted pair. (next slide)

Asymmetric Digital Subscriber Line (ADSL)

- Reserve lowest 25 kHz for voice, known as POTS (plain old telephone service). The voice is carried only in the 0 to 4 kHz band; the additional bandwidth is to prevent crosstalk between the voice and data channels.



(a) Frequency division multiplexing

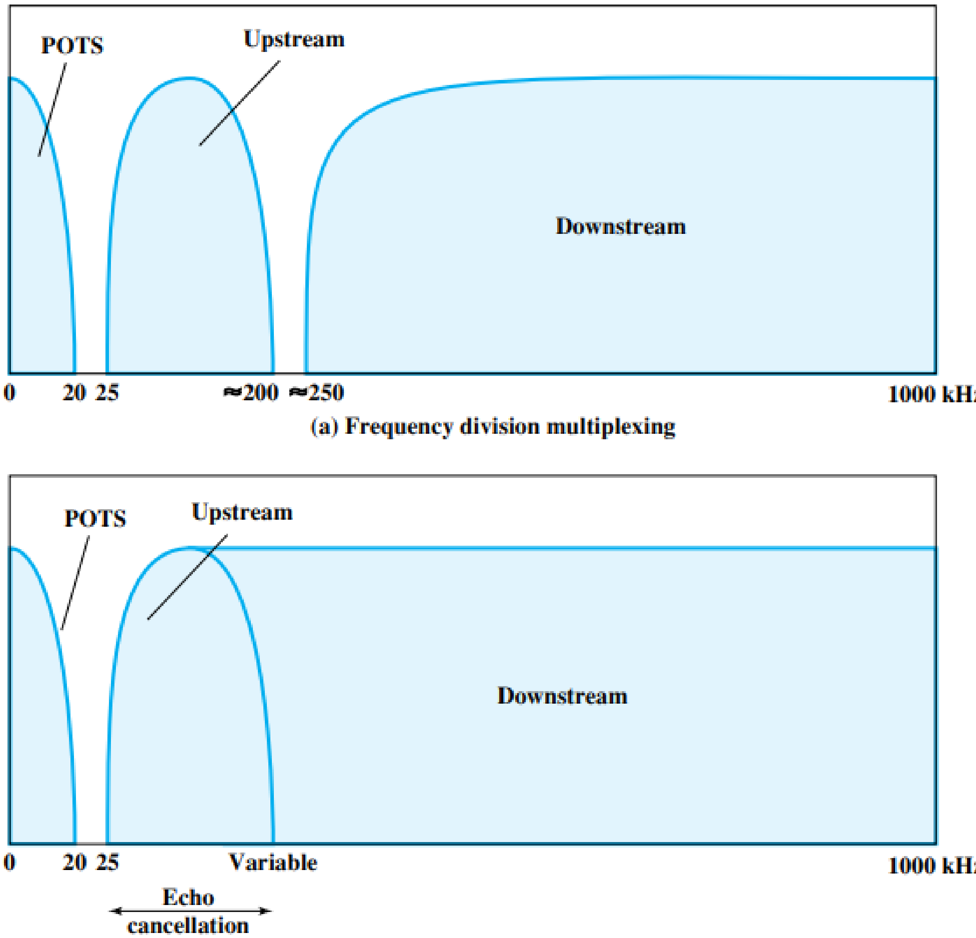


(b) Echo cancellation

Asymmetric Digital Subscriber Line (ADSL)

- Use either echo cancellation or FDM to allocate two bands, a smaller upstream band and a larger downstream band.

  (Echo cancellation is a signal processing technique that allows transmission of digital signals in both directions on a single transmission line simultaneously)



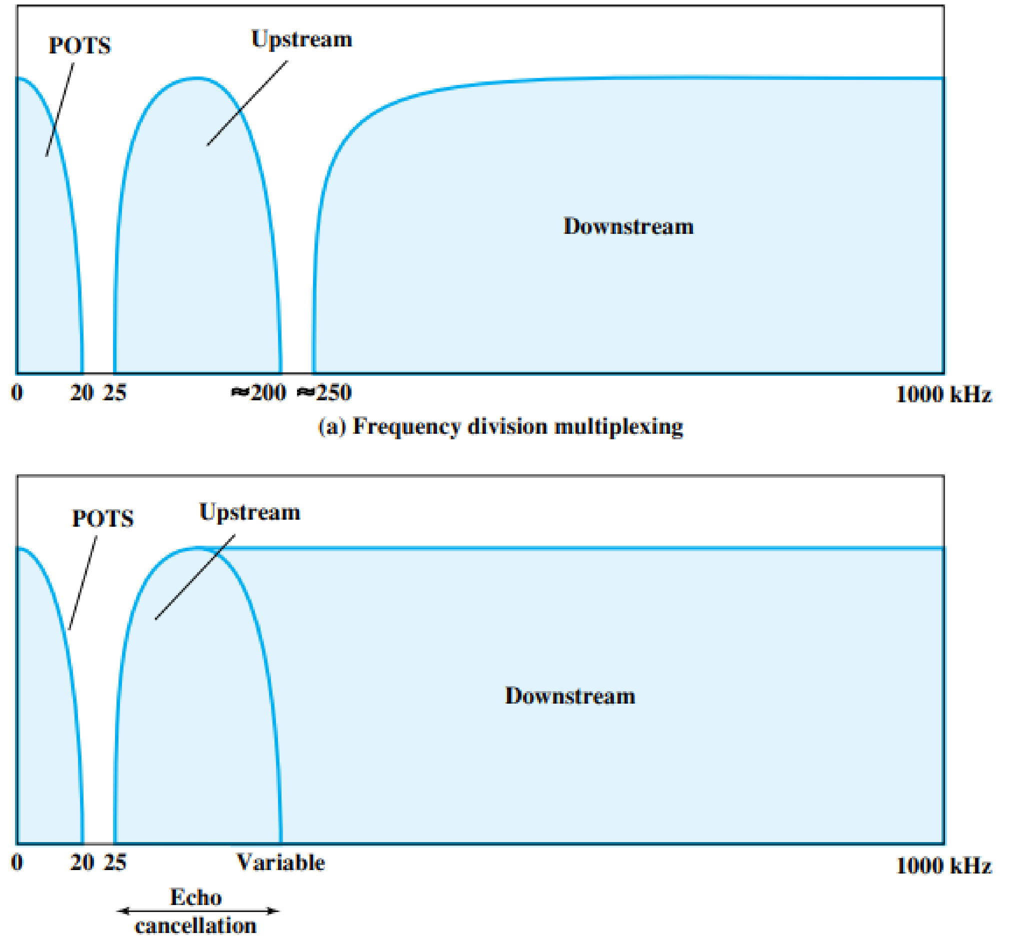
(a) Frequency division multiplexing

(b) Echo cancellation

Asymmetric Digital Subscriber Line (ADSL)

- Use FDM within the upstream and downstream bands.

- In this case, a single bit stream is split into multiple parallel bit streams and each portion is carried in a separate frequency band.



(a) Frequency division multiplexing

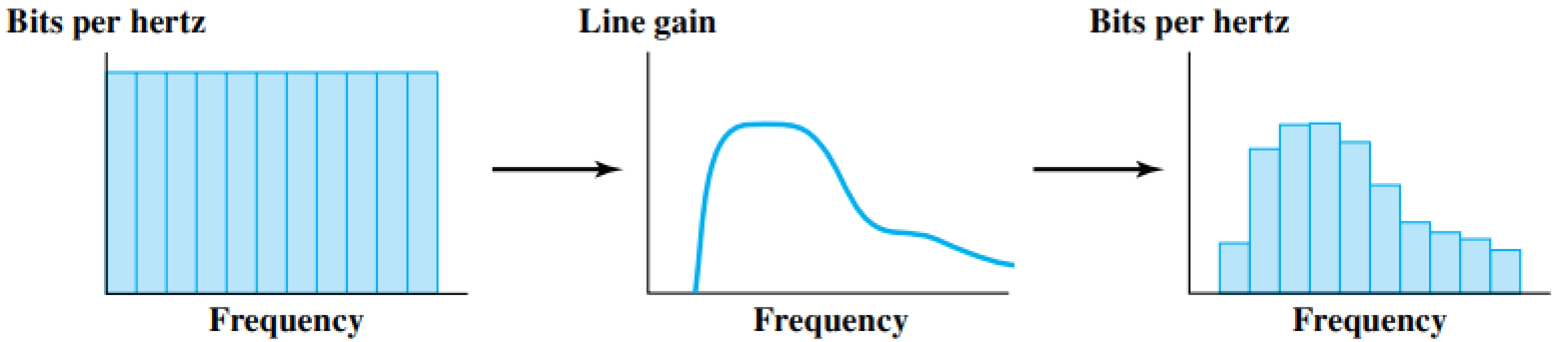
(b) Echo cancellation

Discrete Multitone (DMT)

- Discrete multitone (DMT) uses multiple carrier signals at different frequencies, sending some of the bits on each channel.
- On initialization, the DMT modem sends out test signals on each subchannel to determine the signal-to-noise ratio.
- The modem then assigns more bits to channels with better signal transmission qualities and less bits to channels with poorer signal transmission qualities.

Discrete Multitone (DMT)

- The available transmission band (upstream or downstream) is divided into a number of 4-kHz subchannels.
- Each subchannel can carry a data rate of from 0 to 60 kbps.
- there is increasing attenuation and hence decreasing signal-to-noise ratio at higher frequencies.



References

- Old slides by Prof. Gihan Dias and Dr. Sulochana Sooriyaarachchi

- Data and Computer Communications by William Stallings